

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Real-Time Systems Aug 2025 to May 2026

- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Cut experiment setup effort **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making every experiment sweep reproducible from config.
- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.

Turito | Software Engineering Intern, Low-Latency Systems Jun 2022 to Oct 2022

- Slashed p95 CPU **22%** by profiling hot paths and removing per-packet allocations, payload copies, redundant lookups, and **lock contention** in shared state.
- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Prototyped a **Rust** + **QUIC** media path serving about **8,000** lightweight subscribers with about **24%** lower startup time.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Lowered network egress **14%** by removing redundant media-path processing and simplifying routing and payload handling.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.
- Cut performance-investigation time about **40%** by building tooling on Rust workers, **Python** aggregation, **SQLite**, Flask, and Chart.js.

TECHNICAL SKILLS

Languages: C++17, C, Rust, Python, SQL

Performance: profiling, AVX2/SIMD, multithreading, lock contention analysis, cache-aware memory layout, microbenchmarks

Systems & Networking: Linux, TCP, QUIC, WebSockets, RTP/RTCP, Tokio, CMake, pybind11

Data: HNSW, PostgreSQL, SQLite, Redis

PROJECTS

Proxima, Vector Search Engine | C++17, AVX2/SIMD, HNSW, pybind11

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.
- Cut traversal memory **75%** (51 MB to 12.8 MB) with **int8** AVX2 traversal plus exact float re-ranking and cache-aware node reordering.

TCP Sliding Window Simulator | Python, raw sockets, Linux

- Implemented handshake, sliding window, cumulative ACKs, and retransmission over raw sockets with live packet-state visualization.

OPEN SOURCE

- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Advanced Algorithms, Computer Networks, Distributed Computing, GPU Programming, Database Systems, Machine Learning

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)